

Yeager, Evan

Design and Purpose Statement for the Construction of an Electromagnetic Railgun

3/15/23

Executive Summary

An electromagnetic railgun is a linear motor device that uses electromagnetic force to accelerate a projectile to incredibly high speeds. It works by using two parallel metal rails, with a conductive armature (usually a metal slug) placed between them. When a large amount of electrical current is passed through the rails, it creates a powerful magnetic field that propels the projectile forward, along the rails, and out the end of the device.

Railguns are used for a variety of research purposes, such as testing materials under extreme conditions and studying the effects of high-speed impacts. They can also be used for military applications, such as launching projectiles at high speeds for missile defense or for striking targets at long ranges.

This particular model was designed to use a 4-stage capacitor bank that stores a large amount of charge and releases it all at once. Capacitors were secured from eBay (originally from Lawrence Livermore National Laboratory), they are each rated at 210 microfarads of capacitance and 22,000 Volts, making for a total system energy (at full charge) of 200 kilojoules. The device is to utilize a compressed air injection system to accelerate the armature to the required initial velocity before rail contact to ensure the rails and armature would not crater and/or weld upon contact. A rail enclosure was constructed using fiberglass and epoxy composite (Garolite G-10/FR4) and two 3-foot copper rails. The capacitor bank was to be charged using a custom-built power supply with an output of 0.5 Amps at 22,000 Volts and a gasoline generator. Projected energy efficiency was ~1.25% based on previous research, but the problem of determining final velocity is incredibly multivariate and chaotic, so it was determined that final armature velocity could only be calculated experimentally and creating an estimate would be a futile exercise.

While the primary purpose of the device is for hobbying purposes, it was to be designed with the ultimate goal of using it for materials science, aerospace, pulsed power supply design, and/or circuit architecture research. Some of the potential uses include: i) Orbital debris impact simulation ii) Optimization of capacitor bank architecture for energy efficiency iii) Investigating the relationship between armature material and energy efficiency

Legally speaking, an electromagnetic railgun is not a firearm of any kind. US Code Title 18 Section 921 defines a firearm as "any weapon (including a starter gun) which will or is designed to or may readily be converted to expel a projectile by the action of an explosive." This particular device uses no explosive of any kind. Rather than using a propellant, this device uses high current.

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